

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 February 2026
Team ID	LTVIP2026TMIDS81316
Project Name	TransLingua:AI-Powered Multi-Language Translator
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input Interface	a) User enters text to be translated b) User selects source language (or auto-detect option) c) User selects target language
FR-2	Automatic Language Detection	a. System automatically detects input language when auto-detect is enabled b. Display detected language to the user
FR-3	AI Translation Generation	a) Send user input to Gemini API b) Generate translated output in selected target language c) Display translated text clearly
FR-4	Swap Languages Feature	a. User can swap source and target languages b. Automatically update translation accordingly
FR-5	Dynamic Language Support	a) System supports multiple world languages b) Language list is dynamically configurable
FR-6	Error Handling	a. Validate empty input fields b. Display user-friendly error messages c. Handle API errors gracefully

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application must provide a simple and intuitive interface suitable for all users.
NFR-2	Security	API keys must be securely stored using environment variables. User inputs and outputs must be handled safely.
NFR-3	Reliability	The system must consistently generate accurate translations without crashing.
NFR-4	Performance	Translation responses should be generated within 3–5 seconds depending on input length.
NFR-5	Availability	The application should maintain high availability during active usage.
NFR-6	Scalability	The system should support multiple users simultaneously.